Overview

Team Name: Team Taken
Game Title: *Taken*

The game is powered by a **custom C++ engine** developed entirely from scratch, featuring:
- **Entity-Component-System (ECS)** architecture for efficient runtime updates
- **graphics pipeline** 2D texture and animation rendering
- **Custom editor overlay (ImGui)** for real-time debugging and scene editing
- **Procedural map generation** with automatic door linking and safe-spawn checks
- **Physics & collision system** with rollback recovery
- **Custom GUI system** (Mute / Exit buttons)
- **Font rendering pipeline** using FreeType and OpenGL
- **Audio framework** (FMOD integration planned for future milestones)

Demo Controls and Usage

Keyboard Controls
Key	Action
W / A / S / D	Player movement
E	Use Heal/Burrow Abilities
F	Interact with Enemies/Puzzles
Right Click	Fire Projectiles
F1	Toggle ImGui Editor Overlay
F2	Toggle Custom GUI (Mute / Exit)
F3	Toggle rendering (layering)
F4	pauses game (layering)
F6	Minimising/Maximising window
F7	disables gui
F8	disable Physics debug outline
F11	Toggle Physics Debug Mode
L	Toggle light
R	Mutation Reduction when in debug

```
| **P** | Disable/Enable Paintbrush for Room Editor      |
| **Shift + Left Click** | Allows Multi-Select and Multi-Delete  in Room Editor     |
| **ESC** | Loads pausemenu and how to play when playing  |

## **Cheat Codes**

0: On / Off God Mod
1: Set ability to default
2: Set ability to heal
3: Set ability to burrow
4: Set ability to projectile
5: Set mutation charge up to 0
6: Set mutation charge up to 100
7: Set the player hitpoints to 100
8: Set the player hitpoints to 0

## **Demo Flow**


1. **Launch the App** - Loads MainMenu scene
2. **Play option** - Loads directly into the main Labyrinth map
4. **Quit Game** - Exits game
5. **Press F1** - Opens the ImGui-based EditorOverlay for scene editing and inspection.
6. **Press F2** - Displays the custom in-game GUI with *Mute* and *Exit* buttons.
7. **Enter Play Mode** - The player can move and collide throughout the map, stop resets them back to original position.
8. **Press ESC** - Opens up a pausemenu that pause the game , and shows how to play along with options to Quit,Resume game

---

## **Team Roster**

| Name | Role | Responsibilities |
|------|-------|------------------|
```

Jethro Sung	Tech Lead / Engine Architect	ECS framework, editor overlay, resource & serialization systems, procedural map generation.
Tan Wei Liang Terril	Physics & Collision Champion	Collision detection, physics rollback, physics debugging.	Debug console editor
Woh Kye Le	Graphics Champion, Programmer	Rendering, RoomEditor	Minimap
Low Jian Lin	Audio Lead, Audio Champion	Font, Audio
Sng Swee Yong Dillon	Design Lead, UI Champion	Logic for Game Obj, UI docking/playstop	SceneManager/Mainmenu
Lim Zhi Jie	AI Champion, Programmer	Implementation of AI Logic, Enemies	pauseMenu

---

Engine Overview

The *Taken Engine* provides:
- **Language:** C++17
- **Graphics:** OpenGL (GLAD + GLFW)
- **UI:** ImGui (Editor) + Custom GUI System (Game)
- **Audio:** FMOD (integration-ready)
- **Physics:** AABB collision with rollback safety, broad, narrow phase
- **Math Library:** Custom Vector2 / Vector3 / Matrix3x3
- **Resource System:** JSON-based serialization with prefab instantiation
- **Well Crafted Maps:** Level Design done via Editors that allow designers to fully implement their desired maps and puzzles

**How to Play (For now) **
- Roam the labyrinth
- Enter doors to navigate to different room
- Enemy Ai can shoot projectiles, heal and burrow
- basics puzzles implemented for different rooms

**How to Play (final/future implementation) **
- Roam the labyrinth, pick up weapons, defeat enemies and absorb their powers!
- Strategically make use of these powers to solve puzzles around the map that will allow you to advance further

- Certain abilities can be used at certain areas/mobs only, so beware!
- Make sure to purify at the pods along the way, or suffer a gruesome death.
- Defeat The Warden in order to win!

Built With:

* ###### C++
* ###### glad
* ###### glm
* ###### glfw
* ###### imgui
* ###### Fmod
* ###### freetype

© 2025 DigiPen Institute of Technology - Team Taken.
All rights reserved.